

Date	23 July 2026
Project Name	SB-STOCKS
Maximum Marks	2

Technology Stack

S.No	Technology Layer	Technology / Tool	Purpose / Justification
1	Frontend Framework	React.js	Building dynamic, responsive UI with component-based architecture
2	Frontend Styling	Bootstrap / Tailwind CSS	Rapid UI development with pre-built responsive components
3	State Management	Redux	Centralized state management for complex application state
4	HTTP Client	Axios	Making API requests from frontend to backend
5	Charting	Chart.js	Displaying stock price charts and portfolio analytics
6	Routing	React Router DOM	Client-side navigation between pages
7	Notifications	React Toastify	User-friendly toast notifications for actions
8	Backend Runtime	Node.js (v16+)	Server-side JavaScript runtime for API logic
9	Backend Framework	Express.js	Lightweight web framework for RESTful API development

10	Database	MongoDB	NoSQL database for flexible schema and fast queries
11	ODM	Mongoose	Object Data Modeling for MongoDB schema validation
12	Authentication	JWT	Secure token-based authentication
13	Password Hashing	bcryptjs	Encrypting user passwords securely
14	Environment Config	dotenv	Managing environment variables securely
15	CORS	cors	Enabling cross-origin requests
16	Version Control	Git & GitHub	Source code management
17	API Testing	Postman	Testing APIs
18	Code Editor	VS Code	Feature-rich editor

Technology Stack Justification

The MERN stack (MongoDB, Express, React, Node) was chosen because it allows for full-stack JavaScript development, creating a cohesive environment for developers. MongoDB offers schema flexibility and handles JSON naturally, which fits well with Express and Node.js. React provides a highly responsive UI, which is critical for real-time stock trading applications.